

Design a Perceptron Network for implementing Logical AND gate considering bipolar inputs and target.

Perceptron Learning Rule

$$f(x) = 1 \quad >0, \quad 0 = 0, \quad -1 < 0$$

Epoch= 1

Inputs			Target	Net Input	Calculated Output	Weight Changes $\alpha=1$			New Weights		
x1	x2	b	T	y_{in}	y	$\Delta w1$	$\Delta w2$	Δb	w1(new)	w2(new)	b(new)
				$y_{in} = x_1 * w_1 + x_2 * w_2 + b$		αx_1	αx_2	αt	$w_1(old) + \Delta w_1$	$w_2(old) + \Delta w_2$	$b(old) + \Delta b$
1	1	1	1	$= 0 + 0 + 0$ $= 0$	0	$= 1 * 1 * 1$ $= 1$	$= 1 * 1 * 1$ $= 1$	$= 1 * 1 * 1$ $= 1$	$= (0+1)$ $= 1$	$= (0+1)$ $= 1$	$= (0+1)$ $= 1$
1	-1	1	-1	$=(1*1)+(-1*1)+(1*1)$ $= 1$	1	$= -1$	$= 1$	$= -1$	$= 0$	$= 2$	$= 0$
-1	1	1	-1	$= 2$	1	$= 1$	$= -1$	$= -1$	$= 1$	$= 1$	$= -1$
-1	-1	1	-1	$= -3$	-1	$= 0$	$= 0$	$= 0$	$= 1$	$= 1$	$= -1$

No change in weights as Target and calculate output is correct

Same weights as previous step

Epoch 1 = Final Weights w1 = 1 w2 = 1 bw=-1

Epoch= 2

Inputs			Target	Net Input	Calculated Output	Weight Changes $\alpha=1$			New Weights		
x1	x2	b	T	y_{in}	y	$\Delta w1$	$\Delta w2$	Δb	w1(new)	w2(new)	b(new)
				$y_{in} = x_1 * w_1 + x_2 * w_2 + b$		αx_1	αx_2	αt	$w_1(old) + \Delta w_1$	$w_2(old) + \Delta w_2$	$b(old) + \Delta b$
1	1	1	1	1	1	0	0	0	1	1	-1
1	-1	1	-1	-1	-1	0	0	0	1	1	-1
-1	1	1	-1	-1	-1	0	0	0	1	1	-1
-1	-1	1	-1	-3	-1	0	0	0	1	1	-1

Target output matching Calculated Output

Epoch 2 = Final Weights $W1 = 1$ $W2=1$ $BW=-1$